

Sprint 6 Artifacts

Team #10

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Sprint 6 Goal

The goal of Sprint 6 is to explore experimental social-awareness features within the XR dashboard while also preparing the system for final stabilization, deployment readiness, and demonstration. This sprint investigates whether controlled nearby-user awareness can be introduced through experimental line-of-sight mapping and experimental proximity discovery, while preserving a privacy-first design through shared stats restrictions, security controls, and ethical review. At the same time, Sprint 6 places major emphasis on refactoring, integration hardening, Niantic Spatial NSDK 4.0 compatibility work, deployment cleanup, packaging consistency, and final test preparation.

Sprint 6 builds directly on the modular dashboard architecture, widget isolation work, safety-oriented HUD behavior, Android integration pathways, and cross-device validation from Sprint 5. The focus now expands in two directions at once: first, evaluating whether tightly constrained social XR concepts can be explored safely and ethically; and second, ensuring the broader platform becomes clean, stable, deployable, and easier to demonstrate. This sprint therefore serves as both an experimental research sprint and a late-stage engineering hardening sprint as the project approaches release readiness.

Sprint 6 User Stories

US-6.1: Line-of-Sight Mapping (Experimental)

As a user, I want the system to experimentally investigate whether nearby users can be associated through gaze or directional awareness so social XR interaction concepts can be explored with strict fail-safe behavior.

Primary Contributor: Brandon Dodge and Asa Maker

US-6.2: Proximity Discovery (Experimental)

As a user, I want nearby presence detection to be explored experimentally through anonymous opt-in discovery so shared XR context can be tested without exposing identity by default.

Primary Contributor: Asa Maker and Zach Severt

US-6.3: Shared Stats Rendering

As a user, I want minimal shared stats to appear only with explicit, mutual consent so limited social XR visibility can exist without sacrificing privacy.

Primary Contributor: Jonn Gott and Josh Dwoskin

US-6.4: System Refactor & Cleanup

As a developer, I want the dashboard core and modular widget architecture to be cleaned up and refactored so the platform becomes easier to maintain and stabilize.

Primary Contributor: Asa Maker and Brandon Dodge

US-6.5: Integration Hardening

As a developer, I want the system's major components to function together more reliably so widgets, permissions, dashboard logic, and device communication remain stable under integrated runtime use.

Primary Contributor: Brandon Dodge and Zach Severt

US-6.6: Proximity Security Controls

As a user, I want proximity-based features to require mutual consent, session-based opt-in, and immediate revoke behavior so nearby social features remain under user control at all times.

Primary Contributor: Jonn Gott and Josh Dwoskin

US-6.7: Ethical & Privacy Evaluation

As a developer, I want the team to evaluate privacy risk, misuse potential, and ethical concerns tied to social AR features so future implementation decisions remain responsible and defensible.

Primary Contributor: Jonn Gott and Josh Dwoskin

US-6.8: NSDK 4.0 Integration & Validation

As a developer, I want Niantic Spatial NSDK 4.0 to function within the project pipeline so its compatibility with the existing XR stack can be validated for future use.

Primary Contributor: Brandon Dodge

US-6.9: Deployment Readiness Cleanup

As a developer, I want the project structure and codebase cleaned up for deployment so deprecated assets, disorganized files, and unnecessary dependencies do not interfere with release preparation.

Primary Contributor: Asa Maker and Zach Severt

US-6.10: Build Packaging & Environment Setup

As a developer, I want build settings, dependencies, and environment setup finalized so project deployment remains consistent across systems and development environments.

Primary Contributor: Brandon Dodge and Zach Severt

US-6.11: Release Testing, Documentation & Demo Prep

As a developer, I want final testing, deployment validation, documentation, and demo preparation completed so the project is ready for handoff and presentation.

Primary Contributor: Brandon Dodge and Asa Maker

Sprint 6 Functional Requirements

FR-6.1 Line-of-Sight Mapping (Experimental) (R26)

Derived from US-6.1

- Investigate gaze-based or directional association of nearby users as an experimental feature
- Prevent automatic identification or unsafe persistent user linking
- Require strict fail-safe behavior when confidence is ambiguous or unsupported
- Ensure this experimental feature can be disabled cleanly without affecting baseline dashboard behavior

FR-6.2 Proximity Discovery (Experimental) (R27)

Derived from US-6.2

- Explore opt-in nearby presence detection using anonymous identifiers as an experimental feature
- Prevent nearby-user discovery when consent is missing, revoked, or unsupported
- Avoid exposing personal identifying information during discovery behavior
- Support graceful fallback when discovery data is unstable, inaccurate, or unavailable

FR-6.3 Shared Stats Rendering (R28)

Derived from US-6.3

- Render only minimal approved shared stats under explicit, mutual user consent
- Restrict displayed information to bounded, low-sensitivity data elements
- Support modular dashboard integration through the existing widget framework
- Immediately clear shared information when consent is revoked or the session ends

FR-6.4 System Refactor & Cleanup (R29)

Derived from US-6.4

- Refactor dashboard core systems and modular widget pathways for greater maintainability
- Reduce technical debt in layout, update, and shared-state handling logic
- Preserve compatibility with current widgets and HUD systems during cleanup work
- Improve code organization to support later deployment and documentation efforts

FR-6.5 Integration Hardening (R30)

Derived from US-6.5

- Strengthen end-to-end integration between dashboard systems, widgets, permissions, and device communication
- Validate that experimental and existing features can coexist without destabilizing the runtime
- Resolve cross-system issues that interfere with reliable multi-component behavior
- Prepare the platform for final cleanup, deployment readiness, and release testing

FR-6.6 Proximity Security Controls (R31)

Derived from US-6.6

- Enforce mutual consent before any proximity-linked social feature becomes active
- Support session-based opt-in rather than persistent or automatic participation
- Allow immediate revoke behavior that clears active proximity-linked states
- Ensure security controls apply consistently across shared stats and nearby-user discovery features

FR-6.7 Ethical & Privacy Evaluation (R32)

Derived from US-6.7

- Evaluate privacy implications of nearby-user awareness and social AR data sharing
- Identify misuse risks tied to gaze-linked or proximity-linked interaction models
- Document ethical concerns and acceptable design boundaries for future feature use
- Use evaluation findings to guide safe implementation limits and feature restrictions

FR-6.8 NSDK 4.0 Integration & Validation (R33)

Derived from US-6.8

- Work toward making Niantic Spatial NSDK 4.0 functional within the project pipeline
- Validate whether NSDK 4.0 can operate alongside the existing XR stack and project structure
- Identify compatibility, dependency, and runtime integration issues introduced by NSDK 4.0
- Document findings to support future development and platform decision-making

FR-6.9 Deployment Readiness Cleanup (R34)

Derived from US-6.9

- Remove deprecated assets, scripts, and outdated project components where appropriate
- Clean up project structure so release-oriented work can proceed more predictably
- Reduce clutter and unresolved legacy elements that may interfere with deployment
- Preserve important runtime functionality while improving overall project maintainability

FR-6.10 Build Packaging & Environment Setup (R35)

Derived from US-6.10

- Finalize build settings, dependency handling, and environment configuration
- Improve consistency of project setup across development systems
- Reduce environment-specific failures tied to packaging or missing configuration
- Prepare the project for more reliable deployment and demonstration builds

FR-6.11 Release Testing, Documentation & Demo Prep (R36)

Derived from US-6.11

- Complete final testing and deployment validation across the prepared project build
- Document project setup, known issues, and usage expectations for handoff

- Prepare demonstration materials and project readiness for presentation
- Ensure the project can be communicated clearly as both a technical system and a deliverable prototype

Sprint 6 Delivered Artifacts

Artifact R26: Line-of-Sight Mapping (Experimental)

Sprint 6 begins experimental investigation into line-of-sight mapping as one possible mechanism for socially aware XR interaction. Earlier sprint work provides the modular dashboard, widget lifecycle structure, failure isolation model, and runtime stability needed to support more advanced feature experimentation. This artifact explores whether nearby contextual association can be inferred through gaze or directional awareness while maintaining strict fail-safe behavior. Because this work introduces ambiguity, privacy concerns, and misuse potential, the team treats it as experimental rather than production-ready.

Contributor: Brandon Dodge and Asa Maker

Artifact R27: Proximity Discovery (Experimental)

Sprint 6 also introduces experimental proximity discovery as a way to investigate whether nearby-user presence can be recognized without exposing personal identity by default. This work extends the project's broader interest in contextual XR interaction while remaining aligned with privacy-conscious design goals. The artifact focuses on anonymous identifiers, opt-in participation, and controlled exploration rather than persistent social tracking. It serves as an early research step in understanding whether limited nearby awareness can be technically feasible without compromising user trust or safety.

Contributor: Asa Maker and Zach Severt

Artifact R28: Shared Stats Rendering

With nearby awareness under investigation, Sprint 6 explores whether minimal shared stat rendering can be safely represented inside the XR dashboard. Earlier work focuses on personal metrics, modular widgets, health-oriented display logic, and stable user-facing presentation. This artifact extends that direction by allowing selected information to be shown only under explicit, mutual user consent. The intent is not broad visibility, but rather a tightly constrained proof of concept for privacy-aware social data presentation inside the modular dashboard framework.

Contributor: Jonn Gott and Josh Dwoskin

Artifact R29: System Refactor & Cleanup

As the dashboard grows in complexity, Sprint 6 includes significant work dedicated to refactoring and cleanup across the system core. Previous sprint work provides the architecture, Android integration pathway, widget model, safety behavior, and fault-tolerance patterns, but continued

expansion now requires stronger code organization and reduced technical debt. This artifact focuses on improving maintainability and making the platform easier to stabilize as additional late-stage features and deployment tasks are layered in. It remains a necessary step toward supporting final integration, packaging, and release preparation on a cleaner base.

Contributor: Asa Maker and Brandon Dodge

Artifact R30: Integration Hardening

Sprint 6 places strong emphasis on integration hardening so that the growing collection of dashboard systems can operate together more reliably. Earlier sprint work adds many individual capabilities, including widgets, health metrics, GPS context, safety behavior, and isolation logic, while the current sprint introduces experimental social-awareness pathways and deployment-oriented tasks. This artifact focuses on stabilizing interactions between these systems so feature growth does not produce runtime inconsistency. The result is intended to be a more unified and dependable platform as the project moves closer to release readiness.

Contributor: Brandon Dodge and Zach Severt

Artifact R31: Proximity Security Controls

Because Sprint 6 begins exploring nearby-user awareness, this sprint also introduces stronger security controls governing how such features can activate and persist. Prior sprint work establishes a privacy-conscious direction, but experimental social functionality now requires more explicit boundaries at the system level. This artifact focuses on enforcing mutual consent, session-based opt-in, and immediate revoke behavior so proximity-linked capabilities remain tightly controlled. It helps ensure that any experimental social AR work stays bounded by security rules rather than operating as an unrestricted dashboard feature.

Contributor: Jonn Gott and Josh Dwoskin

Artifact R32: Ethical & Privacy Evaluation

Sprint 6 includes formal evaluation of the ethical and privacy implications introduced by social AR concepts such as gaze-linked association, proximity discovery, and shared stat visibility. Earlier sprint work primarily focuses on technical capability, user-facing utility, and runtime stability, but these newer features require the team to consider misuse risk and responsible design boundaries. This artifact documents the effort to examine whether certain behaviors should remain experimental, limited, or excluded entirely. It remains an important step in keeping the project aligned with its privacy-first goals as more socially aware features are considered.

Contributor: Jonn Gott and Josh Dwoskin

Artifact R33: NSDK 4.0 Integration & Validation

Sprint 6 includes dedicated work toward making Niantic Spatial NSDK 4.0 functional within the project pipeline. This effort focuses on validating whether NSDK 4.0 can coexist with the existing

XR stack, project structure, and broader dashboard architecture without introducing blocking incompatibilities. Because this integration may affect dependencies, environment setup, and future feature pathways, the team treats it as a focused technical validation task rather than a minor add-on. This artifact represents a direct investment in understanding whether NSDK 4.0 can support future spatial functionality within the project.

Contributor: Brandon Dodge

Artifact R34: Deployment Readiness Cleanup

As the system approaches final-stage preparation, Sprint 6 includes deployment readiness cleanup to reduce friction caused by deprecated assets, outdated scripts, and disorganized project structure. Earlier sprint work concentrates on building capabilities and validating system behaviors, but later-stage work now requires a more disciplined release-oriented cleanup effort. This artifact focuses on reducing clutter, removing obsolete elements, and improving the project's structural clarity so deployment work becomes more predictable. It supports the broader goal of turning the dashboard from a growing prototype into a cleaner deliverable.

Contributor: Asa Maker and Zach Severt

Artifact R35: Build Packaging & Environment Setup

Sprint 6 continues release-oriented work by focusing on packaging consistency and environment setup across development systems. As the project grows, differences in dependencies, configuration, and build settings can introduce avoidable instability when preparing demos or deployment targets. This artifact focuses on finalizing build settings, dependency management, and configuration handling so the project can be reproduced more reliably. It serves as an important bridge between active development and a stable project handoff or presentation build.

Contributor: Brandon Dodge and Zach Severt

Artifact R36: Release Testing, Documentation & Demo Prep

Sprint 6 closes with final testing, documentation, deployment validation, and demo preparation so the project can be presented as a more complete and coherent deliverable. Previous sprint work provides the dashboard's architecture, modular behavior, mobile integration, resilience improvements, and expanding feature set, while this sprint adds both experimentation and late-stage cleanup. This artifact focuses on confirming readiness, documenting expectations, and preparing the system for handoff and demonstration. It marks the transition from primarily internal development work toward outward-facing presentation and release confidence.

Contributor: Brandon Dodge and Asa Maker

Sprint 6 Team Contributions

Brandon Dodge

- Contributes to experimental line-of-sight mapping investigation and broader integration

hardening work

- Leads NSDK 4.0 integration and validation effort as a dedicated individual responsibility
- Supports dashboard refactor direction, build packaging preparation, and release-oriented coordination
- Contributes to final testing, documentation planning, and overall sprint direction as the system moves toward deployment readiness

Zach Severt

- Assists with experimental proximity discovery validation and headset-side runtime testing
- Supports integration hardening across current dashboard and device-linked systems
- Contributes to deployment cleanup and build-environment consistency work
- Helps evaluate project readiness under broader runtime and demonstration conditions

Asa Maker

- Extends technical groundwork for experimental nearby-user features and system-level fail-safe behavior
- Helps lead refactor and cleanup planning across dashboard architecture and shared-state systems
- Supports deployment-readiness cleanup and release-focused stabilization efforts
- Continues strengthening the project's architectural foundation for final-stage integration work

Jonn Gott

- Contributes to shared stats rendering design with a privacy-conscious modular presentation focus
- Works on proximity security controls and consent-governed nearby-user feature boundaries
- Supports ethical and privacy evaluation tied to social AR experimentation
- Helps shape how user-to-user visibility remains controlled, minimal, and consistent with project goals

Josh Dwoskin

- Contributes to shared stats rendering and how mutual-consent information can appear within the dashboard
- Helps define proximity security expectations for nearby-user interaction features
- Supports ethical and privacy evaluation of experimental social-awareness functionality
- Continues contributing to user-facing feature planning as the dashboard moves toward late-stage maturity

Sprint 6 Risks Identified & Mitigations

Risk: Experimental line-of-sight mapping may produce ambiguous or unreliable nearby-user associations

Mitigation: Keep the feature experimental, require strict fail-safe fallback behavior, and avoid automatic persistent identification

Risk: Experimental proximity discovery may create privacy concerns if anonymous detection is not tightly controlled

Mitigation: Use opt-in anonymous identifiers, bound the feature to controlled sessions, and treat revoke behavior as immediate

Risk: Shared stat visibility may create unclear expectations around what information is being exposed

Mitigation: Restrict rendering to minimal approved information and require explicit mutual consent before display

Risk: Social AR experimentation may introduce ethical or misuse concerns beyond purely technical implementation issues

Mitigation: Conduct formal ethical and privacy evaluation and use findings to define acceptable design limits

Risk: NSDK 4.0 integration may introduce compatibility, dependency, or environment conflicts within the current XR stack

Mitigation: Treat NSDK work as a focused validation effort, isolate integration issues, and document blockers clearly

Risk: Late-stage refactor, cleanup, packaging, and release preparation may expose regressions or hidden deployment issues

Mitigation: Pair cleanup and packaging work with integration hardening, testing, documentation, and controlled validation before demo use

Sprint 6 Outcome Summary

Sprint 6 focuses on balancing experimental feature exploration with final-stage engineering hardening. The dashboard already has a strong foundation in modular widgets, health-oriented data presentation, Android-linked context, safety behavior, and system resilience. This sprint builds on that base by investigating experimental social-awareness features in a privacy-conscious way while simultaneously refactoring, stabilizing, packaging, and preparing the project for deployment and demonstration.

The expected outcome of Sprint 6 is a cleaner, more integrated, and more deployment-ready XR dashboard that also begins carefully exploring nearby-user interaction concepts without abandoning its privacy-first direction. By the end of the sprint, the team aims to have experimental line-of-sight and proximity feature research, stronger social-feature security and ethical boundaries, dedicated NSDK 4.0 validation progress, deployment cleanup, packaging consistency, and final testing and demo preparation that collectively move the project closer to a polished final deliverable.